POLAR POCKET WPDS

Pocket-FC-06-M	0	March 18, 2026
WPDS No.	Rev.	Date

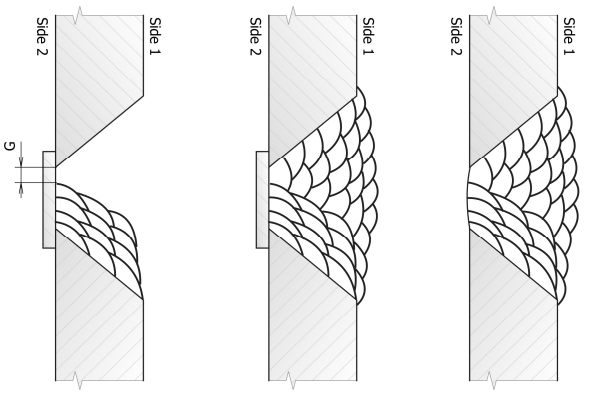
Seaspan Vancouver Shipyards Co. Ltd.

Applicable Standard(s): Lloyds Register - Rules for the Manufacture, Testing and Certification of Materials 2022

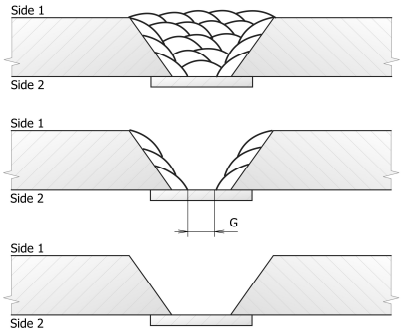
Process/Mode	Electrode (Wire) Classification	Brand Name(s)	Manufacturer(s)
1 FCAW/Semi-Auto(Hand)	**AWS A5.20 E71T-1C/9C-J**	**Dual Shield Prime 71 LT**	**ESAB**

Material Designation	Base material 1	Base material 2	Min. Preheat / Interpass Temp.	**50°C for root and hot pass**, fill/cap to be as per VSY Preheat and Interpass Temperature Requirements for Welding
	EH 36 and all lower grades excluding A,B, D and E (Note 6)	EH 36 and all lower grades excluding A,B, D and E (Note 6)		
Delivery Condition(s)	All except QT	All except QT		
Nominal Pipe Size	**500mm and above**	**500mm and above**	PWHT	**N/A**
Thickness or Dia	**3 mm - 60 mm**	**3 mm - 60 mm**	Max. Interpass	**180°C**
Welding Position	**All positions excluding Vertical down**		Joint Design	**Major/deep weld repairs**

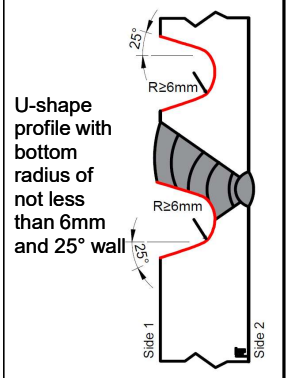
Root gap for ceramic= Target 4-7mm (max 1.5 x T or 11mm, whichever smaller,



TYPICAL 2G BUTTERING/WELDING SEQUENCE



TYPICAL 3G BUTTERING/WELDING SEQUENCE



U-shape profile with bottom radius of not less than 6mm and 25° wall

REPAIR EXCAVATION PROFILE

COMPLETE JOINT PENETRATION	Welding Layer	JOINT TYPE	Back Purge	N/A	Contact Tip to Work Distance	9.5-20 mm
■ **Back-gouged to sound metal**	multi-layer	■ **BUTT**	Backing type	Ceramic/Steel		
■ **Welded onto backing**	One/Two side	■ **CORNER**	Welding Technique	Stringer/Slight Weave	Interpass Cleaning	Grinding and Wire Wheel
☐ Welded from one side without backing	Gun travel angle	■ **LAP**	Max. Bead Width	16mm	Shielding Gas	75% Argon / 25% CO2
☐ Welded both sides w/o back-gouging	Pull w/ slight push on Vu	■ **TEE**	Tungsten Electrode	N/A Ø:	Gas Flow	16-25 LPM
Method of steel preparation	Oxy fuel/Plasma cut | Grinding | Milling	■ **EDGE**	No. of electrodes	1		34-53 CFH

BM Thickness Range mm	Layers / Passes	Position	Electrode Size range (mm)	Welding Process	Current type /Polarity	Consumable	Current (A)	Voltage (V)	WFS (IPM)	Travel Speed (mm/min)	Heat Input kJ/mm
3 ≤ T < 15	Root (Ceramic)	All ex. Vd	1.2 & 1.4	FCAW	CV/DC+	E71T-1M/9M	130 - 250	19 - 25	170 - 300	55 - 130	**1.5 - 2.6**
15 ≤ T ≤ 30											**1.5 - 2.6**
3 ≤ T < 15	Buttering/ Hot/Fill/Cap	All ex. Vd	1.2 - 1.6	FCAW	CV/DC+	E71T-1M/9M	110 - 485	16 - 38	124 - 500	57 - 640	**0.5 - 1.8**
15 ≤ T ≤ 30											**0.5 - 2.4**

Note 1: Heat Input (kJ/mm) = [V x A x 60] / [Travel Speed (mm/min) x 1000]

Note 2: Grind groove joint and adjacent surfaces to bright metal prior to welding to remove all traces of paint, primer, scale, rust, moisture and any other contaminants. Wire brush, grinding to be used for interpass cleaning.

Note 3: Travel angle = 5-10 Pull / slight push should be used for Vertical up welding position.

Note 4: Amps and Volts are to be set by the amp/volt meter at the workpiece.

Note 5: This WPS qualifies both buttering and repairing welds where the gap is less than or equal to 16 mm or 1.5 x thickness, whichever is smaller.

Note 6: As required, gouging may be performed completely through the thickness. Excavated surfaces are to be re-beveled to the original joint configuration. Weld buttering may be performed on one or both faces of the bevel as required.

Note 7: Finished/ground buttered surface shall be successfully examined with MPI or LPI (whichever applicable) prior to application of the final groove weld. Make sure all traces of MPI/LPI consumables are cleaned from the weld joint prior to welding of the main joint.

Note 8: Welding of the normal strength hull structure steel to normal strength hull structure steel (Grade A,B,D and E) using Dual Shield Prime 71 LT is subject to special agreement with Lloyds Register.

Note 9: For the root pass on the ceramic backing, it is recommended to use 1.2mm (0.045") wire size with max. 180 amps.

Engineer Stamp

Reference WPS No.	FC-CS-R-01 (Rev. 2)